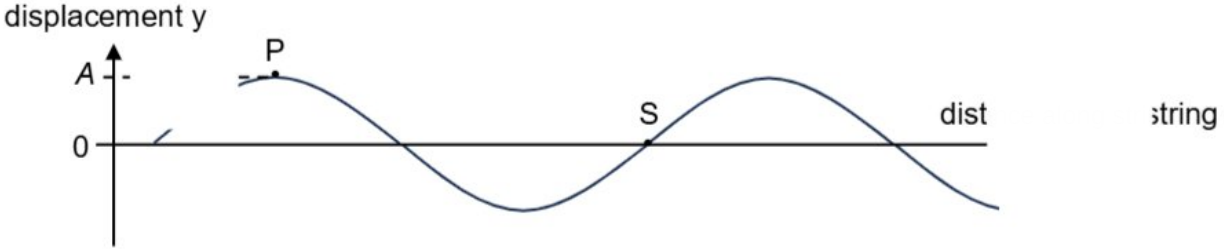
13 The graph shows part of a transverse wave travelling along a string from left to right at time $t = 0$.



The angular frequency of both particles P and S is . The displacements of P and S are y_P and y_S respectively.

Which of the following gives the correct relation between the displacement and time for either P or S and the phase difference between the motion of S and P?

	displacement-time relation of P or S	phase difference
A	$y_S = -A \sin t$	$\frac{3}{4}\pi$
B	$y_P = A \sin t$	$\frac{3}{4}\pi$
C	$y_P = A \cos t$	$\frac{3}{2}\pi$
D	$y_S = -A \cos t$	$\frac{1}{2}\pi$